

BookBonds Product and Build Dossier

Version 3 | September 2026 | Product, experience, systems, and implementation blueprint

BookBonds is a reading community built around real physical books. A member can keep a private reading life, display a personal library, lend a book, give it away, buy an eligible listed copy, or form a Bond by exchanging specific physical copies. Each copy has a persistent platform identity and a collectible 3D book presentation. Readers may visit and build expressive 3D Rooms. A confirmed handoff changes the book's recorded owner or holder, the participants' shelves, its permitted journey, and any eligible rewards. This dossier states the accepted product direction and gives an implementable proposed architecture. It is a planning document, not a claim that any feature, partner, payment protection, authenticity check, or regulatory arrangement is live.

The long-term ambition is one to two million monthly active readers within five years. Here, an active reader completes a meaningful reading, discovery, community, collection, or exchange action during a calendar month. That ambition is a target for experiments and capacity planning, not a forecast.

Scope / accepted concepts, reader experience, platform operations, architecture, and staged delivery

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1 Decisions, language, and boundaries

BookBonds is the platform. A **Bond** is a trade of one or more physical books for one or more other physical books under an agreed exchange; an offer or reservation is not a completed Bond. **Leaflets** are the proposed earned platform currency. A **copy** is one particular physical book; a work or ISBN record does not make it unique. A **passport** is its internal event history. **BondSpine** is the permitted visual account of that one copy's journey; **Bondloom** is the optional Room installation showing a reader's connected reading and exchange history. **Ghost Shelf** is the person's record of books formerly owned and permanently passed on. **Reader's Rooms** are user-built 3D spaces connected to book displays and community. **Keep, Swap, Borrow, and Gift** label permanent, trade, temporary, and uncompensated paths respectively; Gift can be represented as a Keep handoff with no consideration if the user interface makes that clear.

Layer	Source of truth	What people can decorate
Physical book	Copy passport and accepted events	Card frame, display position, binder, Room setting
Agreed handoff	Frozen agreement snapshot and delivery legs	Reveal animation and receipt appearance
Leaflets	Double-entry journal	Statement presentation and reward catalog
Real money	Selected licensed payment providers and reconciliation	Checkout presentation, never a simulated balance
Personal reading	Member-controlled reading entries	Goals, Yearbook, lists, and display choices

Product constraints accepted in discussion: actual copy photos and full cost before agreement; explicit availability and return terms; owner separate from holder; NFC optional; a tag tap is evidence rather than ownership; both required physical handoffs and confirmations before settling a Bond; public history by consent; meaningful reading participation without listing a book. An optional NFT or external ledger remains a later decision. The internal passport and transaction event store remain authoritative unless a future governance plan explicitly changes that boundary.

2 Product principles and audiences

A person should understand a book offer in one glance and a transaction in one screen. The collectible styling creates desire to explore, while condition, owner, holder, obligations, cost, and due dates remain readable. The room and game are enjoyable even for someone who never trades. Member-made notes add humanity but do not certify authenticity. The system must explain uncertainty instead of converting claims into facts. Every important action has a text and keyboard path alongside the 3D presentation.

The main user modes are: a newcomer seeking the next read; a cataloging reader; an active lender; a local or shipping trader; a collector with valuable editions; a circle host; a Room builder; and an authorized partner. One member may move between modes. Avoid a collector-only onboarding path: wishlist, reading log, circles, public Rooms, and eligible gifts or loans provide an entrance for people

without books to trade.

3 Site map and first-use flow

Public site: landing page, how Bonds work, browse actual listed copies, public profiles and Rooms, safety and pricing, partnership information, help, sign in. Reader app: Home, Explore, My Shelves, Bonds and Loans, My Room, Messages, Reading and Circles, Book Passports, Leaflets, Account and Settings. Persistent navigation should keep adding a book and checking pending obligations close at hand. Signed-in Home first shows an offer awaiting a reply, a shipment to prepare, a return due, or a dispute requiring evidence, then discovery and community.

First visit sequence: preview a real interactive sample Room and inspect an example book without an account; join through an accessible secure sign-in method; choose a reading goal; add a wishlist title or physical copy; decide privacy; see one immediate next step. An account does not require a tag, paid wallet, public profile, or full inventory. Provide a guided checklist only until the reader has taken their first useful action. If they skip listing, the app still supports circles, reading log, and Room visits.

Reader settings include profile, handle, avatar, bio and language; security and sessions; discovery radius, shipping and edition preferences; per-book and per-shelf audience; Room draft and published visibility; local meeting preferences and private addresses; notification frequency; blocks; accessibility, motion, sound and 3D quality; Leaflet statement, payment provider links, account export, and deletion request. Show what is public from each setting and require reauthentication for sensitive changes.

4 Book catalog, unique copies, and listing workflow

Represent a work, its edition or format, and the member's physical copy as separate records. ISBN commonly helps identify an edition or format and can prefill bibliographic data; it is not a per-copy serial. Each physical copy receives a platform-issued persistent copy ID. A tag, if used, receives its own ID. Catalog corrections can improve shared metadata while preserving individual copy photos and history. Support unknown ISBNs, older books, imports with nonstandard codes, multiple cover variants, and signed copies.

The listing wizard accepts a barcode scan or manual title search; lets the reader select the right edition; records actual front, back, spine, and flaws; shows condition examples; captures documented special features separately; then chooses Keep, Swap, Borrow, or Gift terms. For Swap, the owner sets acceptable authors, wishlist targets, condition, format, and geography. For Borrow, specify due date or 'return expected' with notice terms. For paid Keep, item price, taxes or fees, delivery charges and refund rules must be displayed before consent. The user saves a draft, previews the public card, and publishes. Photos of a signature or first-edition page are labeled evidence supplied by the owner, not authentication by BookBonds.

Availability states cover unlisted, available, reserved, on loan, in transit, unavailable, and archived. A listing may change while no accepted agreement locks it. Once accepted, preserve the offered copy ID, condition evidence, material terms and cost in the agreement snapshot; subsequent edits apply to the live listing without silently changing the accepted transaction. Prevent one physical copy from being

promised in two active commitments.

5 Smart shelves, binders, and relationship views

Every member has automatic relationship shelves derived from accepted copy and transaction events. **Current** holds copies owned and held by the member. **Away** shows copies still owned but lent to someone else. **Borrowed** shows copies in the member’s possession that belong to another owner. **Pending** shows reserved outgoing copies and incoming agreement previews. **Ghost** contains past-owned copies after permanent transfer, with the member’s own snapshot and approved live journey link. **Borrowed History** retains returned loans without implying past ownership. A book can also sit in user-designed binders, display shelves, themed collections, or a Room; those arrangements do not create a second physical copy.

Event	Sender or lender view	Recipient or borrower view
Swap accepted	Existing copy stays recorded to sender, appears reserved and pending	Incoming preview and agreement receipt
Bond fully settled	Former copy enters Ghost; received copy enters Current	Same with the two copies reversed
Loan received	Lender sees Away; owner unchanged	Borrower sees Borrowed and agreed return terms
Loan returned	Copy returns to lender’s Current	Borrowed History receives a record
Gift completed	Donor receives Ghost	Recipient receives Current

Users can reorder and rename their own binders, choose public or invite-only shelves, and make individual books less visible than the shelf. An empty wishlist slot represents a wanted edition or theme, never an actual available copy. Ghost entries are visually distinct and cannot be offered by a former owner. A member returning to ownership of a former copy triggers Homecoming: new ownership period, preserved past Ghost snapshot, and a consent-filtered journey.

6 Bond Finder, discovery, and matching

Search filters include title, author, edition, format, language, location radius, shipping, offer type, condition, documented features, and price ceiling. Results should expose actual available physical copies, optionally grouped beneath a work record. Wishlist alerts respond to compatible nearby or shippable listings without revealing exact home addresses. Discovery Deck is a small, optional daily set of real copies that mixes direct matches and exploration. Readers can dismiss or tune suggestions.

Bond Finder evaluates reciprocal preferences, wishlists, conditions, distance, shipping willingness and currently available copies. Each suggestion explains the match and estimated obligations. A suggestion does not reserve stock. Open to Bond lets the owner state acceptable offers; Shelf Interest allows a consenting owner to see demand for an unlisted book, with privacy thresholds. Bond Bundles include

multiple books on each side, with independent delivery legs and one frozen agreement. Bond Rings, a later-stage three-person circular trade, require all three people to accept every leg and failure rule before any reservation. Never silently turn a one-to-one request into a larger ring.

7 Bonds, gifts, purchases, and the Bond Table

The Bond Table is a shared visual negotiation space. Each member places a specific 3D copy book with its attached card, actual condition evidence, edition, and availability. Both can propose, counter, remove a copy, or change shipping and pickup terms. Every material counteroffer invalidates the previous acceptance. Before accepting, each person sees the complete agreement: what they send and receive; who pays what; shipping or meetup method; deadlines; photo evidence; loan or gift terms if the path differs; and the issue window. On acceptance, freeze a receipt version and reserve the copies.

A practical Bond state model is Draft -> Offered -> Countered or Accepted -> Reserved -> Preparing -> Dispatched or Meeting -> Delivered -> Recipient confirmations -> Settled. Branches include Withdrawn, Cancelled, Timed out, Disputed, Return requested, and Resolved. Each transition records actor, timestamp, prerequisites, notices, and an idempotency key. A mailed two-book Bond has two delivery legs. Each book can progress independently; final settlement waits for both agreed receipts and any required issue period or documented resolution. A local exchange also requires each participant's confirmation. A single tag tap or a timer does not complete the other person's obligation.

After settlement, owner and holder records update, Current and Ghost views recompute, public permitted BondSpines receive a new segment, Rooms show the new state, the agreement receipt marks completion, and eligible pending Leaflets become available. If a leg fails, pause settlement, preserve the earlier owner records and agreement, gather evidence, and make the decision and appeal route visible. The animation may celebrate a completed Bond only after settlement.

A Gift is a permanent transfer with no book expected back; an open-ended loan is still a Borrow with a return expectation. Paid Keep is a permanent sale or transfer under a separate payment path. These choices must be explicit when people accept so a free handoff cannot become an accidental loan and a loan cannot become an accidental gift.

8 Loans and Reading Routes

A loan retains one recorded owner and gives possession to one current holder. Terms include condition at handoff, date or notice rule, permitted renewals, return method, shipping obligations, optional deposit if ever supported, reminders, damage process, and who can see the loan. On receipt the borrower confirms the book and sees it in Borrowed; the owner sees it in Away. On return the owner confirms receipt; the borrower gains a Borrowed History record. An overdue loan enters a case process without silently changing owner or charging an unagreed fee.

Reading Routes let a book circle circulate one actual copy among consenting people. The queue has current holder, next reader, reading window, skip and pause controls, handoff evidence, and a final return destination. Queue changes are visible to participants; each individual custody change follows the appropriate handoff confirmation. Shared Chapters allows a circle to discuss the same title while

several different copies travel their own independent routes and BondSpines.

9 Shipping protection, disputes, and settlement

Support local meetup and mailed paths, with clear obligations before acceptance. A private shipment record may store label, tracking, dispatch photos and delivery evidence; public views omit addresses and detailed routes. Show the complete real money cost and refund rules, including packaging, postage, coverage and service fees where relevant. If paid transfer is offered, select a provider and a permissible flow of funds before describing it as escrow. BookBonds should not represent a payment as protected by a service it has not contracted and implemented.

Issue types include nonresponse, wrong book, undisclosed condition, damage, lost shipment, late loan, duplicate claim and payment dispute. A case captures the original listing photos and agreement, dispatch and receipt evidence, messages relevant to the dispute, provider references, requested remedy, response deadlines and notices. Outcomes may include continue, amend with both parties' assent, return, replace, cancel, refund under published policy, compensation if available, or adjudicated settlement. Parties see what happened and how to appeal. A mailed issue must not cause automatic transfer simply because tracking says 'delivered.' The protection workflow applies equally to tagged and untagged books.

10 Copy passport and BondSpine

The copy passport stores append-only accepted events and evidence references: registration, initial ownership claim, NFC pairing, offers, accepted handoffs, loan start and return, confirmed permanent transfers, correction, dispute, and tag retirement. Its display state is a projection of those accepted events. Public BondSpine is a readable visualization of only the portion approved for publication. A trade adds a connection; a loan draws a return loop; a disputed or unconfirmed stretch is marked uncertain. The visible card, former-owner Ghost snapshot, optional NFC portal, Room shelf and Atlas all point to the same copy identity.

Earlier holders may contribute optional Journey Notes or Reader Imprints and Sealed Margins for a later reader. The recipient chooses whether to reveal a sealed note. Personal contributions have moderation, spoiler controls and withdrawal; removing a public personal note does not delete the factual audit event. Ghost Echoes notify a former owner of a new journey milestone only if privacy settings allow it. The Atlas maps opt-in activity coarsely; Shelf Portals link approved public Rooms; a Journey Exhibit or Passage Keepsake curates fully consented stories. Neither feature reveals a private holder or precise route by default.

11 NFC identity and offline transfer

Every registered physical copy may exist without a tag. If opted in, a pre-numbered kit records batch, assigned account, shipping and tag state. A member scans an unused label, chooses a copy, confirms pairing and placement, and may photograph installed evidence. Proposed tamper-evident bookplates need physical testing across paper, laminated covers and valuable editions; an adhesive-free path remains available. The proposed 30-plus-digit platform descriptor is a platform-assigned identifier separate from any manufacturer's hardware UID; its printed short reference aids support. It is not a secret credential.

Scanning a tag opens a copy-specific web passport and authenticated actions appropriate to the viewer: Register; Start handoff; Accept receipt; Claim offline transfer; Repair; Replace; Report lost or removed tag; and Request a record correction. The app authorizes each action after sign-in. A new holder who receives a book offline can create an account and submit a claim; recorded owner or holder approval, or an evidence-based staff decision, controls the update. The previous history remains. If a tag is damaged or removed, retire that tag and pair a new tag to the same copy; do not reset the passport. Secure chips may provide a fresh authenticated response per tap, but even that cannot alone prove the chip is attached to the original book or who holds it between scans.

12 Cards, binders, and game rules

The collectible is the specific book. A card is its consistent presentation and readable identity: a compact 3D book within a frame in the binder, a full 3D object on a Room shelf, and the same copy on the Bond Table. Pulling the book closer reveals its front, spine, back and neutral page edges; its attached card unfolds edition, condition photos, documented features, available terms, cost and permitted journey. Never render a signature, wear, or edge color as verified from a generic catalog cover. Readers can collect binders, wishlist slots, earned styles, Reading Yearbook pages, Book Quests and Passage Keepsakes.

Keep four visual signals separate: bibliographic edition facts, documented copy features, confirmed and published journey, and earned personal cosmetics. A foil edge or animated frame indicates a cosmetic, not a rare edition or authenticity. Reading and genuine, completed exchange activities can unlock treatments under rate limits and review. No paid randomized book packs, fabricated scarcity, speculative prices or reward for merely listing hundreds of books. Offer animations must be short, optional and skippable; the full receipt remains accessible while animation plays.

13 Reader's Rooms and the 3D builder

A Reader's Room is a freely designed, visitable 3D scene. Start from a blank canvas or editable study, sunroom, archive, bookstore, train car, rooftop or fantastic library. Place walls, floors, shelves, lights, chairs, plants, artwork, character objects and a reader's registered 3D books; drag, rotate, scale, recolor and light objects. Provide snap guides, undo, autosave, draft and published versions, mobile controls and a private/visitor preview. A room can be a designed place or small story, and does not need to copy the member's real home.

In Visit mode, select a shelf book, lift it, rotate it, inspect its attached card, follow a permitted BondSpine, enter a circle, or request an allowed exchange. In Decorate mode, arrange shelves and props, tune ambience, choose an opening camera position and publish a new version. Current books, borrowed books, wishlist placeholders and Ghost books have visibly different spatial treatments and actions. A pending book remains in the recorded holder's room with a reserved ribbon until completion. A Ghost projection cannot become an active listing. Room visitors receive an audience-specific scene: invited and public guests see only the books and notes shared with that audience.

Create an object catalog and later a creator workshop for original furniture, scenes and approved trigger/action behaviors: a lamp changes lighting, a drawer reveals a permitted shelf, a doorway enters a circle Room. User-made props have licensing, moderation, versioning and attribution. Opt-in room and prop remixing carries creator credit; it does not duplicate an original owner's physical-copy record. Shareable room links, camera snapshots, moderated guestbooks and small themed build challenges support discovery. A Room script or prop may change presentation and navigation; ownership, passport events, Leaflets and payment actions remain platform-governed. Bondloom can weave a visitor-safe picture of the builder's confirmed reading and book journeys.

For implementation, use a common book mesh with per-copy image textures and a reasonable format-dependent thickness approximation, plus shared room assets in a compact runtime format such as glTF. Loading can progressively upgrade textures and simplify light/shadow work on modest devices. Provide full structured card and list views, keyboard control, reduced motion and optional sound. A photographed title page or owner-written note may be inspectable with rights and consent; an unlicensed full book text does not appear as a room feature.

14 Reading, circles, and reader-led discovery

Reading log states are Want to read, Reading, Finished, Paused and Did not finish. Progress, dates, reviews and notes are optional and individually private or shareable. Finished it, pass it on offers Keep, Swap, Gift, Borrow or keep on shelf; a completed read never obliges circulation. Reading Routes curate a sequence of books or themes that people can follow with their own copies; circle Reading Routes coordinate one book's actual custody. Circles support small group membership, hosts, schedules, prompts, spoiler-aware discussion and optional shared reading Rooms. Maintain clear moderation and participant controls.

Home surfaces human recommendations from circles, public shelves and reader-built paths alongside availability-aware suggestions. Community Shelves may later work with bookstores, libraries, cafes or reading clubs, with explicit custody, hours and partner responsibilities. The public Atlas, Journey Exhibits and Shelf Portals connect stories to actual books or permitted Rooms while protecting private holders. Reading Yearbook is an opt-in annual record of reads, handoffs, circles, highlights and approved memories. The goal is a durable reading habit as well as circulating inventory.

15 Leaflets currency and reward economy

Leaflets are a named in-platform reward unit with a familiar account experience: Available, Pending, Held and Spent; a chronological statement; transaction IDs and linked events; a rewards catalog; redemption receipts; a correction request; and understandable terms. Proposed earn sources include confirmed Bonds, genuine completed pass-ons, timely loan returns, reading streaks with restraint, and moderated community participation. Cap and review repetitive events, self-reported activity and coordinated fake exchanges. Release exchange awards only after genuine settlement; disputed awards remain pending or held under published policy.

Potential uses are card and binder treatments, Room items and character objects, visual chat styles, selected physical trinkets, subscription discounts, and partner e-book entitlements. Basic messaging, reading, safety, shelf use and ordinary exchanges must remain available without Leaflets. Each item shows Leaflet cost, any cash due, supply, eligibility, expiry if applicable, delivery, refund terms and support owner before redemption. Only show partner e-books or physical merchandise when rights, territory, inventory, production, fulfillment and funded unit economics exist.

Use balanced ledger postings with accounts for issued awards, user pending, user available, user held, redemptions, reversals and platform obligations according to the selected accounting design. An award event carries source, policy version, unique idempotency key and case link; a posted journal entry is corrected by a compensating entry, never overwritten. Admin grants, freezes, releases and reversals require role, reason, threshold and approval. Reconcile issued, redeemed, outstanding and estimated fulfillment exposure daily. The reader sees a simple statement; Finance sees balanced postings, exception queues and policy simulation. An in-platform reward statement does not imply bank deposits or insured funds.

Launch with earned, noncash, nontransferable rewards usable within controlled platform and partner arrangements if supported by review. Selling Leaflets for cash, peer-to-peer transfers, cash-out or broad merchant redemption remain separate future legal and product decisions. Rewards policy changes should be dated and visible; unexpected devaluation would damage trust. Payment and regulatory specialists must classify the actual chosen arrangement before money-like behavior launches.

16 Partner e-books and other rewards

A partner e-book is a digital access entitlement, not an NFC-tagged physical copy. The record stores partner, title, territory, access method, terms, issuance and support status. The reader sees a Digital Library entry and may add the work to reading log, wishlist and eligible Room display. Resale, borrowing and gifting are available only if the license explicitly permits them. Do not confuse a redeemed entitlement with recorded physical ownership.

Redemption flow: confirm region, member eligibility, partner supply and current price; place a Leaflet hold; request the entitlement idempotently; settle the hold only after a verifiable fulfillment result; otherwise release it and explain why. Reconcile partner invoices and fulfillment reports to BookBonds entries. Partner portal users see only their approved catalog, redemption activity, support cases and contract scope. Partner listings and co-branded shelves require moderation and inventory rules.

17 Account creation, login and security

Offer a passkey path and accessible account recovery; a secure fallback may support people without passkeys. Verify a contact channel, manage devices and sessions, and prompt fresh authentication for address edits, security changes, sensitive redemptions, copy recovery and high-risk handoffs. Notify the member after recovery and authenticator changes. Rate-limit login and claims, monitor bot and credential abuse and provide self-service sign-out from other devices. A scanned NFC identifier or forwarded invitation link opens a route to a request, never an authenticated session or ownership transfer.

Use authorization on every server operation: resource owner, permitted audience, participant status, staff role and assigned case must all match. Staff and partners use stronger login and scoped access; temporary privilege elevation expires. Media uploads require validation and content moderation where appropriate. Protect addresses, recovery material, NFC tag mappings and private evidence with appropriate encryption, storage separation and retention. Keep an incident response path for compromised accounts and delayed handoffs.

18 Reader settings, editing and consent

Settings group	Self-service controls
Identity and security	Handle, public name, avatar, email, passkey, recovery, sessions, export and deletion request
Reading and discovery	Preferred formats, languages, interests, radius, shipping, wishlist limits and alert cadence
Shelves and copy cards	Binder structure, display order, photos and listing drafts, per-book audience, offers and archive
Room builder	Draft, undo, publish version, lighting, prop placement, book placement, preview as visitor
Bonds and loans	Delivery preferences, open requests, return reminders, receipts, dispute submissions
Leaflets and payments	Statement, pending/held explanation, redemptions, issue report, linked payment provider
Community and accessibility	Message requests, blocks, circle alerts, motion, sound, contrast, text and 3D quality

Changes to drafts and personal display can be undone or versioned. Changes to an active listing update the live listing and may require fresh acceptance from anyone with an offer. Changes to a confirmed agreement use an amendment or cancellation path. A member can turn off public travel details and delete their personal contributions under published retention rules while the factual audit remains governed by its own policy. User-facing copy should say why a change is unavailable, pending review, or locked by a transaction.

19 Portals and role model

Reader portal covers books, shelves, Room, transactions, history, privacy and account. Partner portal manages contracted rewards and fulfillment. Staff work is divided into Support, Trust and Safety, Content Moderation, Copy and Tag Operations, Finance Operations, Finance Approval, Audit, and Platform Security. The product owner has broad oversight and configuration rights but high-impact actions retain attributable review. Role permissions specify data fields, geographic scopes, case scopes, allowed transitions and monetary thresholds; staff do not gain access to every private message or address merely because they have an admin login.

Role	Typical capability	Sensitive boundary
Support	Resolve assigned cases and request evidence	Cannot mark a user's receipt confirmed or directly edit Leaflet balance
Copy Operations	Review tags, duplicate claims and repairs	Ownership corrections require evidentiary workflow
Moderator	Quarantine unsafe content and hear appeals	Public note removal does not rewrite handoff events
Finance Operator	Reconcile and propose grants or corrections	Cannot approve own exceptional entry
Finance Approver	Review policy-limited adjustments	Decision and journal remain auditable
Audit	Inspect records and exports	Read-only operational access
Security Admin	Roles, incident response, timed emergency access	Independent audit and review of emergency actions

20 Admin control center

The admin product is a deep operational workbench. A global search accepts member ID, title, copy ID, NFC serial, Bond, shipment, Room, reward, Leaflet journal entry, payment provider reference, report or partner. Each object page shows current state, linked objects, full permitted timeline, original evidence, staff decisions, policy version, queues and available actions. Saved operational views include overdue loans, unresolved one-sided Bonds, suspicious tag claims, reward reconciliation failures and Room content reports.

Admin domains: member access and recovery; catalog and physical copies; NFC kit production and pairing; Bonds, loans, gifts, paid Keep and shipping; passport and BondSpine visibility; Leaflet issuance and redemption; partner e-book inventory; Rooms, creator props, circles and messages within case policy; discovery ranking, notifications, community programming, reward rules, geography, subscriptions, integrations, service notices and security operations. Authorized admins can inspect, configure, pause, repair, publish and review every product domain through product controls. Critical

changes do not rely on ad hoc database edits.

The change composer previews before and after, affected readers, linked copies, user-facing notices, accounting effects, rule version and reversibility. Normal edits, such as a catalog typo or a featured Room placement, can publish after reason capture. Sensitive changes, such as owner corrections, large Leaflet grants, bulk messages, policy thresholds, account reinstatement after fraud or payment repair, require case evidence, step-up authentication and independent approval according to policy. Bulk operations have dry-run counts, staged rollout, per-item results and a rollback plan. If the early team has one operator, define an external second approver or defer actions that require dual review once real value or member rights are affected.

A read-only View as Reader preview shows what the member sees on a shelf, Room, agreement or message notice, with private fields masked unless access is needed for an assigned case. Acting for a member uses an expressly attributed staff transition and user notice where appropriate. It cannot forge the person's confirmation. The platform security owner can activate time-limited emergency access or pause new offers, tag activation, Leaflet redemption, a payment integration or a region; all emergency actions are logged, reviewed and reversed through controlled restoration. Policy changes have versions, owners, effective dates and simulation results. Audit search includes privileged reads, exports, role changes, denials, overrides and emergency access. The audit trail remains protected from silent alteration, including by a super-admin.

21 Trust, privacy, moderation and abuse

Signal labels must be precise: owner supplied photos; claimed special feature; tag paired; secure scan verified if available; both sides confirmed handoff; claim under review. No blanket authenticity badge derives from a barcode or tag. Public geography is coarse and opt-in; addresses, exact handoff routes, private conversations and loan terms remain restricted. Community moderation supports block, mute, report, evidence request, warning, temporary restriction, takedown, appeal and restoration. Room props and community-created models have publish review and rights-complaint processes.

Fraud controls look for fake Bond loops made for rewards, duplicate copy images, tag re-use or tampering, abnormal claims, account farming, suspicious redemption velocity, refund abuse and harassment. A signal opens proportionate review, not automatic guilt. Define initial member eligibility, age and guardian policy before public social features. Data inventory and retention schedules identify personal notes, addresses, messages, proof photos and factual events separately. The reader controls sharing; a member's deletion request follows law and operational retention rules rather than falsely promising immediate deletion of all transaction evidence.

22 Three financial and historical records

A copy passport, Leaflet journal and real-money provider ledger answer different questions. The passport answers who is recorded as owning and holding a book, and why. The Leaflet journal answers which earned units were granted, reserved, held, spent or reversed. The payment provider answers which real-money charges, refunds, payouts and chargebacks occurred. Connect them by immutable identifiers and reconciliation, not by assuming a cash payment or Leaflet award transfers a book. A correction to one record may trigger reviewed compensating events in others.

At the application layer, all write commands are authenticated, authorized and idempotent; the server validates the required state before accepting a transition. Publish a domain event only after durable commit, using an outbox or equivalent delivery pattern. Projections update shelves, Rooms, notifications, search, passport views and Leaflet eligibility. Retries can reprocess an event without duplicate transfers or awards. If a projection lags, show an updating state while the authoritative record stays available. Payment webhooks require validation, deduplication and reconciliation to provider records.

If a future NFT or public-chain layer is considered, specify whether the token references a physical copy, a platform account claim, a participation memento or another right. Define who can mint and transfer it, whether users need wallets, how custody disputes and mistaken transfers are corrected, how costs are paid, and how the public chain can avoid revealing private people or addresses. A token or on-chain transfer cannot establish current physical possession by itself. Pilot this only if member research shows a real benefit beyond the existing passport, not as an assumed requirement for registering a book.

23 Conceptual data model

Record	Important fields or relationship
Account and access	Account ID, preferences, role grants, recovery, sessions, audience and eligibility
Work and edition	Work ID, edition ID, ISBN if any, authors, publisher, language, cover source
Physical copy	Copy ID, edition link, photos, features, current owner and holder projection, status
NFC tag and kit	Platform tag ID, hardware signal, kit, batch, shipment, pairing and retirement
Shelf and Room scene	Viewer/audience, placement, asset references, draft/live version, prop entitlement
Offer and agreement	Participants, offer versions, accepted snapshot, costs, copy IDs and obligations
Delivery leg and loan	Sender, recipient, shipping/meetup evidence, receipt, return and issue windows
Passport event	Actor, type, source, before/after, evidence, verification level and visibility
Leaflet journal	Journal group, debit, credit, amount, rule version, event key, approver and reason
Payment reference	Provider IDs, charge and refund states, reconciliation status, access scope

Record	Important fields or relationship
Case and evidence	Issue type, assigned role, decisions, review, attachments and audit log
Reading and circle	Progress, dates, privacy, circle members, prompts and route membership
Partner entitlement	Reward SKU, terms, region, redemption and fulfillment reference

Each edition can have many copies. A copy has one persistent ID, many historical owners and holders across time, zero or one active paired tag, and multiple retired tags. One agreement may contain several copies and delivery legs. An individual member may own a copy, temporarily hold another, and display a former copy as a Ghost at the same time. Unique constraints and state transitions enforce these rules in storage and in application logic. Public passport material is a curated projection, never a raw dump of private events.

24 Suggested service and interface boundaries

A web application can serve public browsing, reader account screens, and the 3D Room renderer. Server modules or services own Identity and Permissions, Catalog and Media, Physical Copies and Passports, Bonds and Loans, Payments, Leaflets, Rooms and Entitlements, Community, Notifications, Search, and Admin Cases. Initially a modular server with explicit ownership and one transactional database can reduce distributed coordination. Separate scale-critical rendering, search, media processing and background work behind well-defined boundaries as measured load requires. This is an architecture proposal rather than a required vendor stack.

Client commands include create listing, propose Bond, accept version, mark dispatched, confirm received, report issue, start loan return, submit offline claim, pair or retire tag, place book in Room, redeem reward and publish scene. Each command returns a stable request ID and a current state. Webhooks or polling update long-running workflows; notifications are secondary and cannot be the sole source of required information. Use image storage with signed access, a CDN for public assets, role-aware API responses, a search index of permitted listings, queues for background events, and monitoring for errors and settlement lag. Keep secrets and sensitive evidence outside the 3D scene payload.

25 3D and asset build process

Create a small art and rendering system before commissioning hundreds of props. Define scale, book mesh dimensions, material palette, light budgets, camera behaviors, prop attachment points and accessibility labels. Build one base book mesh with replaceable front, back and spine textures; support a catalog illustration or clearly generic placeholder when no copy photo exists. Actual condition remains inspectable in the owner's photos. An edition may inform approximate thickness but the visual should never claim to reconstruct the exact object's undocumented flaws.

The first prototype should prove three linked motions: pick a 3D book from a Room shelf, rotate and open its readable card, and place that same copy into a Bond offer. Then prove the physical handoff

state changes the two users’ Rooms and shelves while the former owner keeps a Ghost. Prototype keyboard, touch and reduced-motion behavior at the same time. Optimize texture sizes, instance repeated props, progressive loading and compact 3D asset delivery. Measure on a modest phone as well as desktop. Later creator tools can support licensed user-made props and constrained approved behaviors with sandboxing and asset review.

26 Engineering, design and operational build process

Discovery: interview readers who swap, borrow, collect and only read; observe current local and mailed exchanges; test what ‘Keep’, ‘Swap’ and ‘Borrow’ mean to people; study photo capture effort and privacy expectations. Design: map jobs and critical paths, prototype the card and Room at real mobile sizes, test offer comprehension and cost visibility, map empty and failure states, and write a service blueprint for support. Technical foundation: define IDs, constraints, event types, agreement state transitions, privacy fields, authorization matrix, double-entry invariants and provider boundaries before core UI implementation.

Implement vertical slices: (1) account to actual copy listing; (2) discovery to accepted Bond to complete two-sided local handoff; (3) loan and return; (4) 3D book to Room display and Bond Table; (5) shipping leg and dispute; (6) Leaflet earn and spend; (7) optional NFC tag activation and offline claim. Each slice includes admin case visibility, instrumentation and a tested failure path. Deploy behind flags by geography or cohort. Invite a small pilot, examine support cases weekly, revise policies and copy, and only then broaden distribution. Document operational runbooks before promising protection, partner rewards or staff intervention at scale.

A practical core team covers product and UX research, interaction/3D design, web and backend engineering, security/privacy ownership, trust and support operations, and later payment/partner expertise. Exact staffing and dates depend on funding, scope and whether licensed components are bought or built. Use a clear definition of done: working success path, keyboard path, permission checks, audit entries, recovery, instrumentation, and support instructions.

27 Milestones and stage gates

Stage	Reader-facing result	Gate before expansion
Foundation	Accounts, work/edition/copy model, listings, Current and Ghost, privacy, basic discovery	Actual-copy identity, accessible listing and server authorization proven
Exchange core	Bond Table, agreement snapshot, two-sided local confirmation, loans and returns, case intake	No duplicate reservations or premature owner changes in realistic pilots
Spatial collection	3D book card, editable first Room, binder, visitor preview, gradual effects	Mobile performance, faithful copy state and accessible equivalent proven

Stage	Reader-facing result	Gate before expansion
Community and rewards	Circles, notes, Reading Paths, Leaflets journal and controlled cosmetics	Reward abuse limits, balanced postings and case workflow tested
Shipping and NFC	Provider-backed shipping, tagged passports, kits, offline claims, replacements	Policy and fulfillment pilots, tag material tests, loss/damage response
Partner scale	E-books, physical rewards, subscriptions, community shelves, creator props	Signed rights, funded unit cost, moderation, support and reconciliation
Advanced network	Atlas, Bond Rings, deeper Rooms, optional NFT study	Evidence of reader value, privacy, technical and legal feasibility

Stage gates describe dependencies, not calendar promises. The 3D book and a genuinely editable Room belong in the product identity; the first release may use a compact prop catalog and one scene per person while still delivering the central spatial interaction. Ship shipping or payment protection only once the provider, liability and dispute process are real.

28 Verification and acceptance scenarios

An accepted swap stores the exact copies, photos, cost and both parties' agreement version. Duplicate requests cannot reserve a copy twice. Two shipments have distinct leg statuses; confirmation of one leg does not settle both. When both complete, exactly one owner/holder transition per copy is recorded, Ghost and Current update, and the Leaflet award posts at most once. On a disputed leg, neither celebration nor irreversible ownership award runs until resolution. Loan receipt changes current holder, not owner; return restores the correct shelf and leaves Borrowed History. A tag scan can open a passport but cannot accept an offer or claim a book by itself.

A Room visitor sees only public books and permissions; the owner sees private return information only in private mode. Moving, remixing or duplicating a decorative book prop never creates a second registered physical copy. A Leaflet redemption that fails fulfillment releases its hold; a refund creates new entries and leaves a trace. Admin corrections require the configured reason and approval and do not rewrite earlier facts. Test permission boundaries between users, staff roles and partners; concurrency, retries, offline users, screen readers, reduced motion, bad networks, and payment webhook replay. Verify that the final text equivalent permits a complete exchange without entering the 3D world.

29 Economics, growth and learning

The five-year user ambition requires repeated local value and a functioning shippable supply, not only downloads or collectible visuals. Pilot one dense geography such as Dallas-Fort Worth if recruitment and operations support it. Seed quality inventory through member shelves, circles and interested partners; show nearby matches and genuinely shippable choices; cultivate enough return visits from reading and Rooms that members stay active between handoffs. Use opt-in invitations, public Rooms and Yearbook shares to bring in new readers without exposing private book histories.

Measure activation to first wishlist or listing, actual available copies per active reader and region, match-to-offer, offer-to-agreement, agreement-to-confirmed exchange, time-to-completion, loan return health, repeat reading, Room visits leading to a book inspection, and 30/90-day meaningful retention. For operations measure cases per hundred handoffs, resolution age, fraud rate, tag kit cost, lost-shipment rate, Leaflets outstanding, partner redemption cost and support load. Define MAU explicitly and report reading/community actions separately from transactions. Experiments should improve reading and trust rather than maximize reward farming or message volume.

Economics model must include user acquisition, moderation, storage and 3D bandwidth, catalog/photo processing, shipment support, NFC materials and postage, refunds/disputes, rewards liability, partner license, and customer care. Every reward promised has a funded expected redemption cost. Expansion decisions compare actual completed exchanges and reader retention with incremental operational cost. No numerical revenue, cost or growth projection in this dossier should be treated as validated without pilot data.

30 Open decisions and research backlog

Launch geography and age eligibility; which Keep listings may be free or paid; postage payer, meetup safety and optional coverage; payment and shipping providers; refund, loan deposit and open-ended return policies; first partner e-book contract and rights; initial NFC chip and label material; whether a sticker is sent free or sold; thresholds for Leaflet earn, spend and approval; account recovery and data retention specifics; creator asset licensing; Room hardware floor; search ranking objectives; and whether an external NFT ledger offers value beyond a private audited passport. Decide public journey granularity with members rather than assuming travel history is always welcome.

Prioritize unresolved decisions by risk and dependency: payment legal and custody model before paid protection; edition and copy ID before any listing; agreement state before Bonds; privacy/authorization before Room publishing and Atlas; journal design before Leaflets; tag material pilots before mass mailing; rights and funding before e-book redemption. Record a named decision owner, date, evidence, alternatives and chosen policy for each. Update this dossier when those decisions are made.

31 Accepted feature inventory

The following names and concepts were accepted during planning. This index prevents an individual idea from disappearing when requirements are split across teams. Implementation stage is still subject to validation.

Feature	Role in the product
Bond Finder	Explains feasible reciprocal matches between actual available copies
Open to Bond	Lets a copy owner express acceptable trade terms
Shelf Interest	Optional signal that an unlisted visible copy has interested readers
Bond Bundles	Multiple physical books per accepted trade, with each delivery leg tracked
Discovery Deck	Small optional daily set of real copies to explore
Journey Notes and Reader Imprints	Opt-in human story contributions from people who held a copy
Bond Web	Private or consented view of books and people connected by confirmed handoffs
Handoff Invitations	Recipient-specific invitation for an offline exchange or gift
Reading Yearbook	Opt-in annual account of reading and confirmed circulation
Reading Routes	A circle plan for one actual book moving between readers
Reading Route Queue	Next reader, handoff window, return rules and route recovery
Community Shelves	Managed physical partner locations with defined custody
Collector Care	Careful-handling path for fragile or valuable documented copies
Book Quests	Optional reading and community activities with capped cosmetic rewards
Bond Table	Shared visual agreement space for actual copy books and terms
Bond Rings	Later three-person circular trades with all parties' consent
Living Cards	One specific copy's card evolves through confirmed events and permitted notes
Ghost Echoes	Consent-gated news about a former owner's passed-on book
Book Portal	NFC or QR route to public passport and authorized management actions
Rooms That Live	Room changes reflect genuine current, pending, borrowed and Ghost states
Living Reading Paths	Reader-authored thematic sequences other people can follow
BookBonds Atlas	Coarse opt-in map of permitted journeys and community activity
BondSpine	The visual, consent-filtered history of one physical copy
Sealed Margins	Optional message from an earlier holder revealed by choice

Feature	Role in the product
Homecoming	One copy’s documented return to an earlier owner
Shelf Portals	Permitted connection from a Ghost or journey to a public Room
Journey Exhibit	Editorial stories assembled from consented book journeys
Passage Keepsakes	Export or later physical memento of approved journey material
Shared Chapters	Circle discussion for one title across distinct physical copy histories
Bondloom	Optional Room installation of one reader’s approved reading and book network

The signature throughline is a specific physical copy carried consistently across its 3D book, attached card, Bond receipt, current shelf, Ghost memories, passport, BondSpine, Room, and eligible game progression. Features that aggregate other readers’ data need both publisher consent and data minimization before they are visible.

32 Worked journeys and failure paths

A two-person mailed Bond

Jadyn has Copy A and accepts Edgar’s offer of Copy B. Both inspect condition photos and agree that Jadyn pays his own outbound postage and Edgar pays his. The accepted version freezes both copies, photos, return/dispute terms, and total obligations. Each sender receives a private shipping preparation checklist; neither participant sees the other’s address outside permitted fulfillment. Both cards are Reserved, still owned by their original owners, and visible as Pending previews to the other person. Jadyn posts his tracking; Edgar posts his. Tracking updates each delivery leg and can show a book In transit, without changing ownership. Jadyn receives Copy B and confirms its condition; Edgar must still receive and confirm Copy A. If Edgar files a damage report, the Bond pauses with evidence and a timeline. After a documented return, adjustment with both parties’ agreement, or adjudication, the platform applies only the supported ownership outcome. A clean completion updates two copies once, creates two Ghost snapshots, updates both Rooms and BondSpines according to privacy, and releases one eligible reward per rule. Duplicated carrier webhooks do not repeat settlement.

A no-date loan and return

Jadyn offers a book as Borrow with ‘return expected’ and a specified notice period. Edgar agrees to handling and return postage before receiving it. After handoff confirmation, the owner remains Jadyn, Edgar becomes recorded holder, Away and Borrowed update, and the private card shows return terms. When Jadyn requests return, the app schedules reminders under the agreed notice period and lets Edgar request a practical extension. Edgar confirms dispatch, Jadyn receives and checks condition, then the holder returns to Jadyn and Edgar gets a Borrowed History record. If Edgar is unresponsive, a case opens with agreed terms and evidence; time passing does not turn the book into Edgar’s property.

An offline gift found through an NFC tag

A person receives a previously registered book at an in-person event. They scan the bookplate, see only a permitted public passport, create an account, and select Claim received copy. The form asks how they received it, who may confirm, and whether they can photograph the unique copy or tag. The prior recorded owner or holder receives an invitation to confirm if reachable. Where the prior participant cannot be reached, a staff case compares evidence, last known state and conflicting claims. The claim stays Pending review; neither scanning nor entering the long tag descriptor grants control. If confirmed as a permanent gift, post one correction or handoff event, update owner and holder, link the prior owner to Ghost, and preserve the offline gap as such. If the tag was removed, retire it and pair a replacement without changing the underlying copy ID.

Inspecting a Room and offering a book

A visitor opens a published Room link. The scene renderer asks for the visitor's audience permission and serves only allowed objects and copy summaries. They tap a spine, the 3D book lifts, and a card unfolds to show actual-copy photos, permitted journey, Swap terms and any shipping requirements. The visitor puts one of their own eligible copies on a Bond Table. The owner counters, changing the return method and both postage obligations. Both see the new total before accepting. If the book becomes reserved elsewhere while the visitor is drafting, the table refreshes and invites a new offer. If the visitor cannot use 3D, the same book facts and request workflow exist in a text/card view. After a successful physical exchange, the Room reflects the new ownership and the former owner's Ghost display according to each person's choices.

A partner e-book redemption

A reader spends a displayed number of Leaflets on a licensed e-book. The app checks territory and partner inventory, places a Leaflet hold and requests one entitlement with a unique external reference. A partner timeout leaves the order Pending while reconciliation queries its real result; retrying must not buy the entitlement twice. If the entitlement arrives, the hold settles, a Digital Library item appears and the partner access link is shown. If the partner rejects the order, the hold releases, the statement records the failed attempt and support receives a traceable case. A real-money tax or delivery fee, if ever applicable, would have been disclosed separately before confirmation.

33 Admin control examples and operational runbooks

Wrong recorded owner. Support receives a report and creates a case; the interface shows the copy passport, accepted agreement, photos, handoff confirmations, tag status and linked member notices. The member's current display may show 'Ownership under review.' Copy Operations proposes a correction event with evidence and affected records. An authorized approver sees a before/after preview of Current, Away, Ghost and Room changes. Once approved, the copy projection updates and both participants receive a reasoned notice with an appeal path. The prior record remains visible to authorized reviewers.

Accidental mass Leaflet award. Finance freezes the affected issuance rule, identifies event keys and posts no further awards under that version. The reconciliation workbench lists impacted accounts,

pending versus spent units, partner redemptions and expected cost. A correction plan defines holds, compensating entries, user communication, and whether any redeemed benefit can practically be reversed. The approver reviews totals and a dry run. Individual corrections post balanced journal groups; the original awards remain in the statement with explanations. The policy returns through a versioned release after a verified test.

Unsafe creator Room asset. Moderation quarantines the prop version and sees all published Rooms that reference it. The owner can hide it immediately from visitors while preserving private drafts, notifying affected builders, preserving evidence for an appeal and inviting a safe replacement. Copies and Bonds displayed in those Rooms stay in their authoritative records; the prop action never changes a book's owner or a member's Leaflets. Rights and content decisions use recorded policy and a second reviewer when consequences are severe.

Incident mode. A compromised staff credential triggers revoked sessions and keys, an emergency pause for exposed workflows, a scoping search across privileged reads and writes, notification decisions, restoration from known-good policies and a post-incident review. Break-glass use is time-limited and automatically queued for independent review. Existing Bonds and loans continue to display their obligations even if new acceptances are paused.

34 Data integrity and permission invariants

A registered copy has exactly one persistent copy ID; no more than one active tag pairing; and at any accepted moment one recorded owner and a defined holder or explicit in-transit/unknown custody state. Multiple displays can refer to the copy but cannot create another owned physical copy. A pending incoming card is a right to receive under an agreement, not current inventory. A Borrowed Shelf item cannot be offered by the borrower as owner. A completed Bond references the same agreement version both sides accepted and settles all required legs exactly once. A cancellation or dispute resolution closes reservations and carries an explanation.

A Leaflet journal group balances; event keys make awards and redemptions idempotent; balances are computed; manual grants carry actor, reason and rule version. Real-money totals reconcile to provider references. Every user-facing state has an authorization decision independent of what the client requested. Room render requests filter assets and linked copy details for the specific audience. Staff access to a case leaves a trail; a role grant does not hide earlier staff activity. A tag's long ID is public-facing lookup material, never an ownership credential. Background projections may lag but cannot invent a confirmation. These invariants should become automated checks and incident alarms.

35 Art direction and visual system

BookBonds should feel like a grown-up reading space that occasionally becomes theatrical. Typography uses a readable display serif for book titles and restrained interface type for action, condition and price. Warm paper, deep ink, oxblood, moss, violet and brass accents can distinguish surfaces without encoding important states through color alone. A 3D book is the hero; its card frame can earn foil or edge treatments while verified copy features remain labeled. The standard Room themes are starting materials, not restrictions on imagination. Character items and trinkets add warmth and personality without turning every room into a shop.

Interaction language is tactile: lift, turn, shelve, inspect, offer, send, receive. Use one clear primary action per transaction step, plain language for irreversible changes, and a quiet status cue for reservation. Provide light and dark reading modes, adjustable type, captions for sound-dependent elements, focus indicators, short animation durations and reduced-motion equivalents. Public sharing and cross-room discovery should never require readers to decorate; a minimal default Room still displays books beautifully.

36 Source notes and implementation references

Product choices in this dossier are proposals developed from the BookBonds conversation and the prior dossier. The following primary sources help bound specific technical or security claims; they do not endorse BookBonds or resolve commercial contracts.

The next working artifacts should be a prioritized requirements board with owners and acceptance scenarios, followed by a clickable prototype of the 3D book, Room, Bond Table, shelves, and Leaflets statement. Partner contracts, pricing, legal structures and service promises must be chosen and validated before they are advertised to readers.

- International ISBN Agency: ISBN guidance
- NFC Forum: NFC technology overview
- NXP: NTAG 424 DNA and TagTamper information
- NXP: Secure NFC application note AN12196
- Things Inc.: Rooms
- Khronos Group: glTF 2.0 specification
- W3C: Web Content Accessibility Guidelines 2.2
- NIST: Digital Identity Guidelines SP 800-63B
- OWASP: Application Security Verification Standard
- OWASP: Authorization guidance
- OWASP: Logging guidance
- FinCEN: Prepaid access guidance